Importing Libraries

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
%matplotlib inline
```

1. Reading Images using OpenCV

```
# Read chess images as grayscale
chess16 = cv2.imread("chess16.png", cv2.IMREAD_GRAYSCALE)
chess64 = cv2.imread("chess64.png", cv2.IMREAD_GRAYSCALE)

# Read avocado image as color
avoc = cv2.imread("avoc.png", cv2.IMREAD_COLOR)
```

2. Display Image Attributes

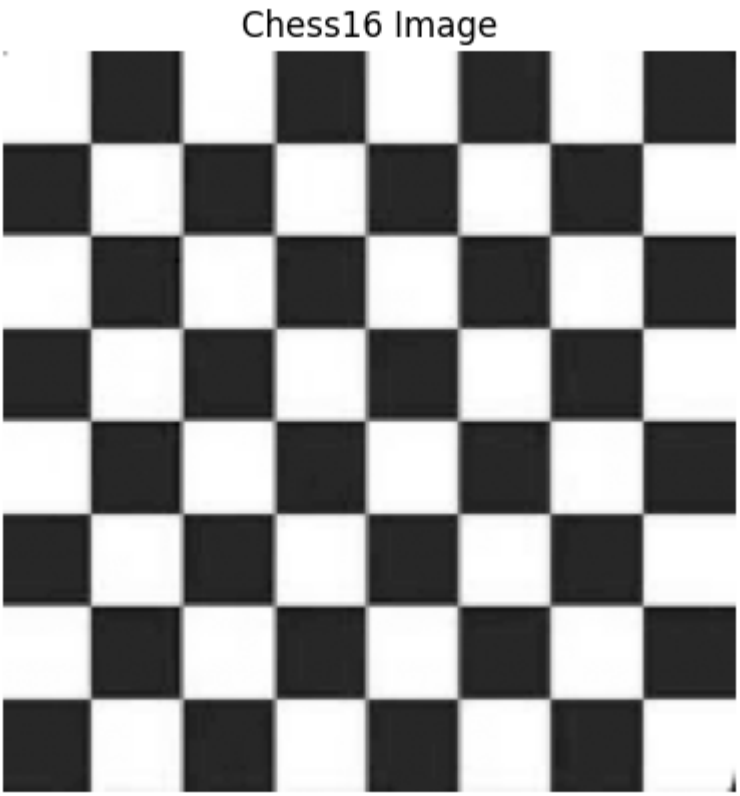
```
print("Chess16 Shape:", chess16.shape)
print("Chess16 Data Type:", chess16.dtype)
print("Chess16 Size:", chess16.size)

print("Chess64 Shape:", chess64.shape)
print("Chess64 Data Type:", chess64.dtype)
print("Chess64 Size:", chess64.size)

print("Avoc Shape:", avoc.shape)
print("Avoc Data Type:", avoc.dtype)
print("Avoc Size:", avoc.size)
```

```
Chess16 Shape: (168, 166)
Chess16 Data Type: uint8
Chess16 Size: 27888
Chess64 Shape: (168, 170)
Chess64 Data Type: uint8
Chess64 Size: 28560
Avoc Shape: (176, 227, 3)
Avoc Data Type: uint8
Avoc Size: 119856
```

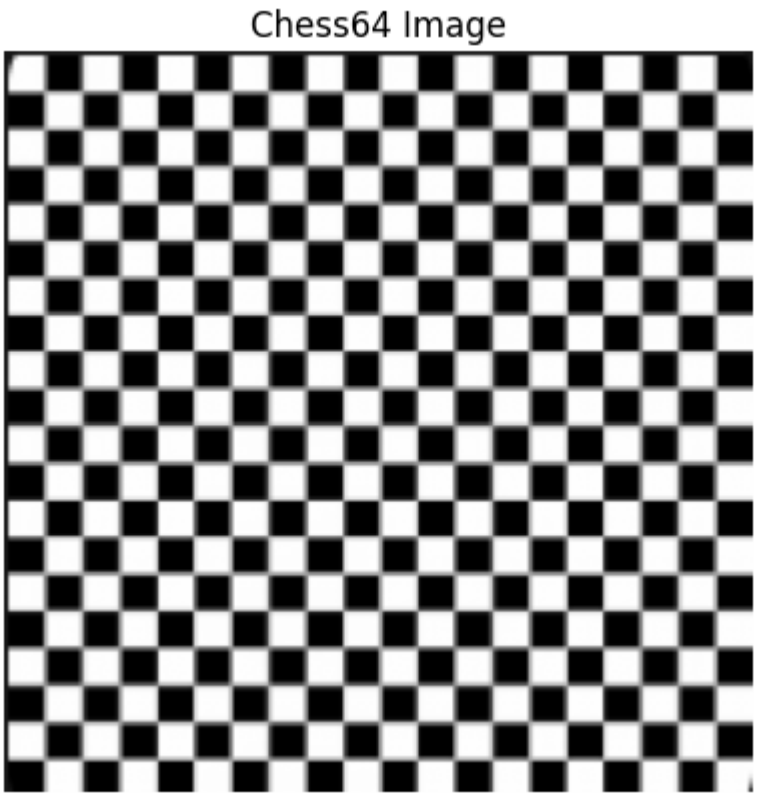
3. Display Images using Matplotlib

```
plt.imshow(chess16, cmap="gray")
plt.title("Chess16 Image")
plt.axis("off")
```

```
plt.show()
```




```
plt.imshow(chess64, cmap="gray")
plt.title("Chess64 Image")
plt.axis("off")
plt.show()
```



```
plt.imshow(avoc, cmap="gray")
plt.title("Avoc Image")
plt.axis("off")
plt.show()
```

4. Working with Color Images

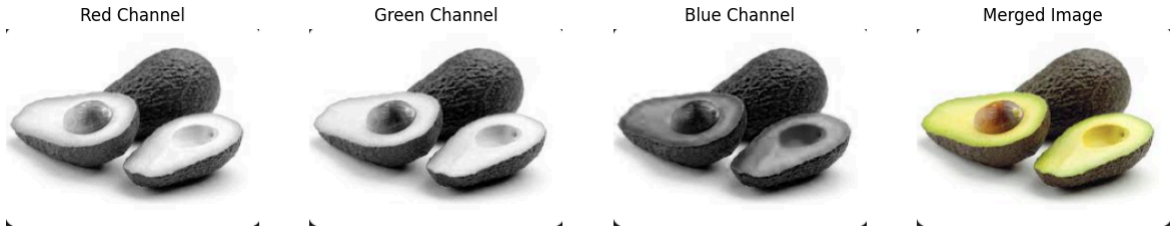
```
print("Shape:", avoc.shape)
print("Size:", avoc.size)
print("Data Type:", avoc.dtype)

plt.imshow(avoc)
plt.title("Avoc Image (Wrong Colors - BGR)")
plt.axis("off")
plt.show()
```

```
Shape: (176, 227, 3)
Size: 119856
Data Type: uint8
```

```
avoc_rgb = cv2.cvtColor(avoc, cv2.COLOR_BGR2RGB)

plt.imshow(avoc_rgb)
plt.title("Avoc Image (Correct RGB)")
plt.axis("off")
plt.show()
```

```
avoc_reversed = avoc[:, :, ::-1]

plt.imshow(avoc_reversed)
plt.title("Channels Reversed")
plt.axis("off")
plt.show()
```

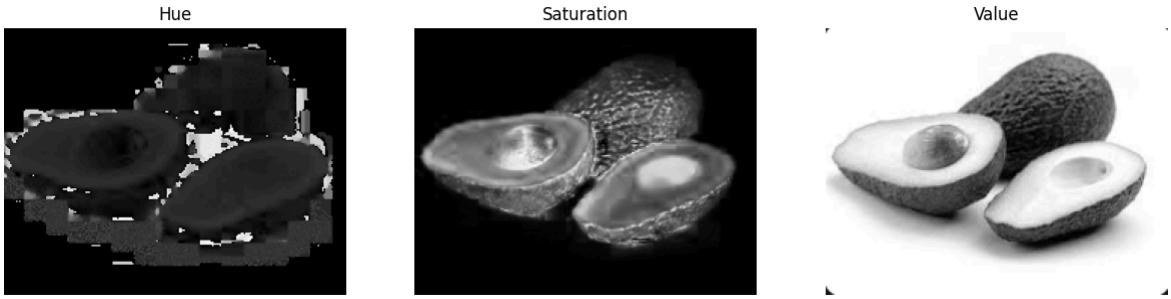
```
# Split channels
b, g, r = cv2.split(avoc)

plt.figure(figsize=(15,5))

plt.subplot(1,4,1)
plt.imshow(r, cmap="gray")
plt.title("Red Channel")
plt.axis("off")

plt.subplot(1,4,2)
plt.imshow(g, cmap="gray")
plt.title("Green Channel")
plt.axis("off")

plt.subplot(1,4,3)
plt.imshow(b, cmap="gray")
plt.title("Blue Channel")
plt.axis("off")

# Merge again
merged = cv2.merge((b,g,r))

plt.subplot(1,4,4)
plt.imshow(cv2.cvtColor(merged, cv2.COLOR_BGR2RGB))
plt.title("Merged Image")
plt.axis("off")

plt.show()
```




```
avoc_rgb = cv2.cvtColor(avoc, cv2.COLOR_BGR2RGB)

plt.imshow(avoc_rgb)
plt.title("BGR to RGB")
plt.axis("off")
plt.show()
```

BGR to RGB

```
avoc_hsv = cv2.cvtColor(avoc, cv2.COLOR_BGR2HSV)

h, s, v = cv2.split(avoc_hsv)

plt.figure(figsize=(15,5))

plt.subplot(1,3,1)
plt.imshow(h, cmap="gray")
plt.title("Hue")
plt.axis("off")

plt.subplot(1,3,2)
plt.imshow(s, cmap="gray")
plt.title("Saturation")
plt.axis("off")

plt.subplot(1,3,3)
plt.imshow(v, cmap="gray")
plt.title("Value")
plt.axis("off")

plt.show()
```




```
avoc_gray = cv2.cvtColor(avoc, cv2.COLOR_BGR2GRAY)

plt.imshow(avoc_gray, cmap="gray")
plt.title("BGR to Grayscale")
plt.axis("off")
plt.show()
```

```
# Increase Hue value
h_new = h + 30

# Merge back
modified_hsv = cv2.merge((h_new, s, v))
```

```
# Convert back to RGB for display
modified_rgb = cv2.cvtColor(modified_hsv, cv2.COLOR_HSV2RGB)

plt.imshow(modified_rgb)
plt.title("Modified Hue Image")
plt.axis("off")
plt.show()
```